

AutoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 2: Model Testing, Validation and Refinement

J.O. • Dr. David M. Simmonds • Evidence through October 1, 2026, 4:01 PM CDT

PRE-TEST WORKING DRAFT — NOT YET SUBMISSION-READY

The frozen factual candidate, representative answer-level benchmark, human leakage disposition, one-time held-out test, same-split baseline, robustness measurement and current result-reproducing notebook are not complete. This document prepares the mandated structure without claiming a test score, threshold verdict or deployment readiness.

1. Summary of Results

No valid model-versus-base factual test result exists at this evidence cutoff. Sixteen completed nested-size training runs (four each for Muse, Gemma, Nemotron and Ornith) used fixed validation prompt IDs and zero test/control rows; they demonstrate token fit, not policy-answer accuracy. An untouched base and a validation-selected winner have not been paired on a locked final benchmark. The result against the planned ten-percentage-point factual-accuracy improvement and safety targets is therefore undetermined—not a pass or fail.

Decision measure	Current observed result	Precommitted target / meaning
Answerable factual accuracy; base versus selected model	Not measured on a shared locked final set	Model improvement at least +10 percentage points versus untouched base
Correct source locator	Not measured on a representative final set	At least 95% on the relevant scored denominator
Unsupported factual claims	Not measured on a representative final set	Fewer than 2% of scored claims
Correct abstention	Not measured; original 30 controls are all answerable	At least 90% on a meaningful unsupported-question stratum

Business translation is pending: report correct supported answers, unsafe claims and appropriate abstentions per 100 relevant cases only after observed denominators exist. Manual 30/30 on simple page/code lookups is an easy-task ceiling, not a same-split comparison or measured analyst time savings.

2. Experimental Setup

Target: a sourced answer to a public Alabama Medicaid policy/provider-billing question, or an abstention when the frozen source does not support one. It is not an autonomous eligibility, payment or clinical determination. A source file, page/section locator, answerability label and source quote define the adjudication record. The deep-learning family comprises locally adapted language models; ALPHIN is the model component, not the name of the overall project.

Pipeline and Protection

The canonical ordinary supervised corpus has 1,999 answerable records and a document-grouped historical 1,684/124/191 train/validation/test split. All 30 original student-authored control items map by source URL plus SHA-256 to historical test documents, with none mapped to clean train/validation; their wording and answer keys are independently authored but their source documents are not independent of the historical test group. The historical 191 have already been exposed as diagnostics and cannot now serve as an untouched final test. A new representative, source-verified control/test specification and immutable membership must be recorded before scoring.

Preparation → pinned base and response-only tokenizer/chat template → nested train subset (N=200/509/750/1,600) → fixed 124-row validation → per-run checkpoint selected by within-run validation loss → candidate freeze → exactly one model/base final evaluation. Preserve document identity, prompt-ID hashes, image/code revision, decoding settings, answer-key/source provenance and selected adapter digest. Training rows or test/control rows must not be introduced by an evaluation step.

Metrics and Business Asymmetry

Primary: human-adjudicated answerable factual accuracy. Secondary: verifiable locator accuracy, unsupported-claim rate, correct abstention on an adequate unsupported stratum, safety routing, general-task regressions and latency. A false confident policy claim is costlier than an abstention that triggers analyst review. Teacher-forced next-token accuracy and loss are training diagnostics, not answer-level validation scores; no cross-tokenizer factual winner may be inferred from them. The earlier notebook reproduces data and legacy diagnostics, not the current 16-run matrix or an unperformed final test.

3. Iteration Narrative

The disciplined experiment loop changed training-set size while preserving the same validation prompt IDs. Every run has a recorded rationale, selected checkpoint and train/validation token metrics in `run_log.csv` and its separate cohort artifact. This is an engineering refinement arc, not a validated answer-level model selection.

Family	Nested sizes completed	N=1,600 final / best step	What the runs establish
Gemma 4	200, 509, 750, 1,600	600 / 400	Token-fit trend within Gemma; selected adapter is not a factual winner.
Muse Glimmer	200, 509, 750, 1,600	600 / 200	Token-fit trend within Muse; pre-amendment September runs excluded.
Nemotron 3.5 Lightning	200, 509, 750, 1,600	1,200 / 400	Corrected-code validation/early stopping; no protected rows.
Ornith 1.5	200, 509, 750, 1,600	1,200 / 400	Corrected-code validation/early stopping; no protected rows.

Failed or Non-Comparable Attempts

Early September Muse experiments were superseded by changed generator/split/training methodology; their metrics are retained as history and excluded from the current matrices. The first Nemotron N=200 attempt failed before training on an unsupported warmup argument. Pre-fix Nemotron N=200 and Ornith N=200/N=509 attempts failed selected-checkpoint finalization when a nested train evaluation cleared scheduled validation. A targeted callback fix was regression-tested, then all eight architecture runs were repeated from pinned bases in fresh directories; no failed attempt is presented as clean evidence.

Within-run checkpoint selection used validation loss, early-stopping patience two and minimum improvement 0.005 for the eight corrected architecture runs. All eight archives have 28 matching host-to-local file hashes. These checks verify training integrity. They cannot replace the Week 8 thirty-error human review, named/countable policy failure modes, a targeted model-quality fix, Week 9 stability check, or a frozen factual candidate. Those tasks remain open.

4. Test Results — Evaluation Not Yet Run

The one-time final test is still locked. A legitimate same-split table requires one preselected candidate, the untouched base, the same fixed test records and decoding settings, a signed answer key and a committed scorer. Do not fill missing measurements with token loss, a single unanswerable example, or the historical 191 diagnostics.

Same locked final test condition	Answerable factual accuracy	Correct locator	Unsafe claim / abstention
Untouched base; fixed generation settings	Not run	Not run	Not run
One frozen validation-selected candidate	Not selected / not run	Not run	Not run
Business thresholds (predeclared)	At least +10 pp versus base	At least 95%	Unsafe claims <2%; correct abstention ≥90%

Confusion-Matrix Equivalent

After the one-time run, publish counts and denominators for answerable cases answered correctly with supported locator; answerable cases answered wrongly or without support; answerable cases unnecessarily abstained; and unanswerable cases correctly abstained versus answered unsafely. Maintain separate answerability strata so a pooled accuracy cannot hide an abstention failure. No cell currently has a measured count; blank is not zero.

Business Units and Test Discipline

Once measured, compare base/model deltas in correct supported answers per 100 answerable cases and unsafe responses per 100 unsupported cases, then test the precommitted pass/fail gates. No time-saving percentage is supported by the retrospective manual lookup proxy. If the first test reveals a defect, preserve and diagnose it; do not tune on those final outcomes or choose a different winner after exposure. Record an honest negative result if the threshold is missed.

5. Error and Robustness Analysis

The Week 8 requirement is a human review of thirty individual validation errors, three to five named business-language failure modes with counts and a targeted change. No verified thirty-case review, model-output error taxonomy or post-change answer-level validation result is present at this cutoff. Sixteen cross-split duplicate-answer groups involving fifty-seven generated records are data-quality flags, not measured model-error counts.

Planned failure mode (not an observed count)	Required diagnostic / possible response
Exception or effective-date mismatch	Inspect source version/date and competing rule; correct data or abstain.
Unsupported confident policy claim	Check citation against exact passage; escalate and enforce scope rule.
Locator cites wrong source or page	Check generated locator, not merely URL presence; repair parsing or retrieval.
Abstention on answerable / answer on unsupported	Human-adjudicate separate strata; revise decision boundary only on development data.
Numeric code/table-cell mismatch	Compare exact source column; quarantine known H2011 “Daily” ambiguity until reviewed.

Robustness Protocol — Not Run

Before the final evaluation, predefine a realistic input shift such as a faithful paraphrase of a policy question, a missing source locator, or a later document revision. Freeze the model and score the unchanged benchmark alongside the perturbation using the same primary human-adjudicated accuracy. Report both denominators and the percentage-point movement; this is an evaluation of robustness, not permission to tune after test exposure. No robustness delta exists yet.

6. Subgroup Performance and Risk

Policy-support risks differ by answerability, document family, exception/date requirements and source vintage. Predefine those segments on the protected benchmark; report per-segment counts, accuracy, unsupported claims, locator correctness and abstention with denominator and uncertainty. A single pooled token-level score does not establish equitable or reliable policy support. There are no verified model-output subgroup gaps to report at this cutoff.

Segment / expected use	Measurement after freeze	Decision if gap appears
Answerable versus plausible unanswerable	Separate factual accuracy and correct-abstention denominators	Block unsupported confident answers; route for analyst review.
Provider manuals versus Administrative Code	Answer/locator accuracy and error counts by document family	Inspect extraction, revision and source coverage.
Simple lookup versus exception/date/multi-rule reasoning	Stratified accuracy and unsupported-claim rate	Limit scope of model-aided drafting until reasoning cases pass.
Current versus later source revisions	Stale-source and date-mismatch failure count	Versioned retrieval and manual approval; retrain only after review.

Risk Framework Applied to This Decision

The National Institute of Standards and Technology AI Risk Management Framework (NIST AI RMF 1.0) offers Govern, Map, Measure and Manage functions. For this use, Govern requires an accountable analyst and source-version owner; Map identifies incorrect policy advice and unsupported claims as harms; Measure requires the answerability and source-family slices above; Manage blocks deployment when factual or abstention gates fail and triggers monitoring/escalation. This is a risk-management mapping, not a claim that the system has passed an audit or meets a specific regulatory classification. Source: <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>

7. Limitations and Deployment Readiness

No factual candidate, final-test score or measured robustness/subgroup result supports deployment. Current public source documents can change; a static adapter is not a current-law source. The original thirty control questions are personally authored and source-verified but are simple and concentrated; the historical test document membership is known, not an independent fresh test. No new student-authored reasoning or plausible-unanswerable set has been supplied; do not have an AI assistant write those questions.

Before final evaluation: human-dispose all sixteen duplicate-answer groups and the near-duplicate question pair; verify or quarantine the H2011 table-column error; author/source-check broader controls including eight to ten plausible unanswerables; commit the new split and score specification; perform Week 8 individual validation-error review and one targeted development fix; test stability on validation and freeze one checkpoint plus untouched base. Then run the protected test exactly once, compute human-adjudicated metrics/robustness/subgroups and execute the current raw-data-to-result notebook. Until then the answer is “not ready for autonomous policy decisions.”

Operating and Monitoring Conditions

Any future pilot must use versioned authoritative retrieval, clear source locators, analyst approval before use, abstention/escalation for unsupported or conflicting guidance, access controls and an audit trail. Monitor factual/locator/unsupported-claim/abstention rates by source family, answerability and date; investigate source-version drift and new document families. Reconsider deployment or retraining if a precommitted safety threshold is missed, source revisions invalidate citations, or subgroup gaps appear. These are proposed safeguards, not implemented or measured runtime capabilities.

8. Repository and Reproducibility

Project repository: https://github.com/porespellar/INAI6986_project_deliverables. The archived earlier-phase notebook is https://github.com/porespellar/INAI6986_project_deliverables/blob/main/notebook/milestone1_reproducibility.ipynb ; it does not reproduce the current 16 training runs or a nonexistent final test. A current raw-input-to-test-metric notebook link must be added after the one-time result exists. The sixteen-row run log and separate 28-epoch/24-epoch matrices are the current training evidence.

AI Use Disclosure

AI assistance: Hermes Agent (GPT-6 Sol 900k) drafted this evidence-bound document and code, cross-checked rubric/weekly requirements and summarized verified local artifacts; earlier local model workers generated supervised examples. No chat assistant authored new independent-control questions or produced final-test scores.

Human authorship/accountability: J.O. conceived and designed the project, selected models/data/metrics, set up local hardware, directed agents, troubleshooted training, chose recovery/quarantine actions and source-verified the original thirty questions and keys. The student must independently review this draft, complete the pending evaluation and accept final responsibility before submission.

Appendix A — Evidence and Outstanding Acceptance Gates

Claim or gate	Authoritative evidence / present disposition
Frozen source and historical split	approved_candidates.jsonl (1,999 answerable records); 1,684 train / 124 validation / 191 already-exposed historical test; document-grouped split.
Original independent set	student-authored-control-set-2026-09-22.jsonl: 30 personally authored and source-verified lookup questions, 30/30 source documents historically test-only; new reasoning/abstention items pending.
Training evidence	results.jsonl: 28 Gemma/Muse epoch records; architecture-training-gates/clean-runs/architecture-clean-run-matrix.json: eight Nemotron/Ornith runs and 24 epochs; run_log.csv: 16 experiments.
Failed attempts	docs/engineering/records/superseded-pre-fix-architecture-runs-20261001.json; incomplete training finalization superseded and excluded.
Leakage disposition	16 duplicate-answer groups / 57 generated records; near-duplicate question pair and H2011 source-column check await human ruling.
Final-test and notebook	Candidate not selected, expanded control not locked, one-time final test unused; current test-result notebook not available.
Assignment scope	Milestone 2 rubric, Parts A-C; Week 7-10 checklists in OneDrive_AUM/Weekly_work/. Due October 25, 2026, 11:59 PM CT; current Canvas wording should be confirmed before submission.

This appendix records exact evidence boundaries. It neither changes the held-out data nor turns a pre-test working draft into a completed Milestone 2 submission.